

Autonomous Obstacle-Avoidance Robot

Based on DFRobot Cherokey 4WD Platform

Team Members | Course Name | Date

Project Overview



Phase 1: Hardware

Assembly following the DFRobot manual basics.



Phase 2: Bluetooth

Manual control testing via iOS app verification.



Phase 3: Autonomous

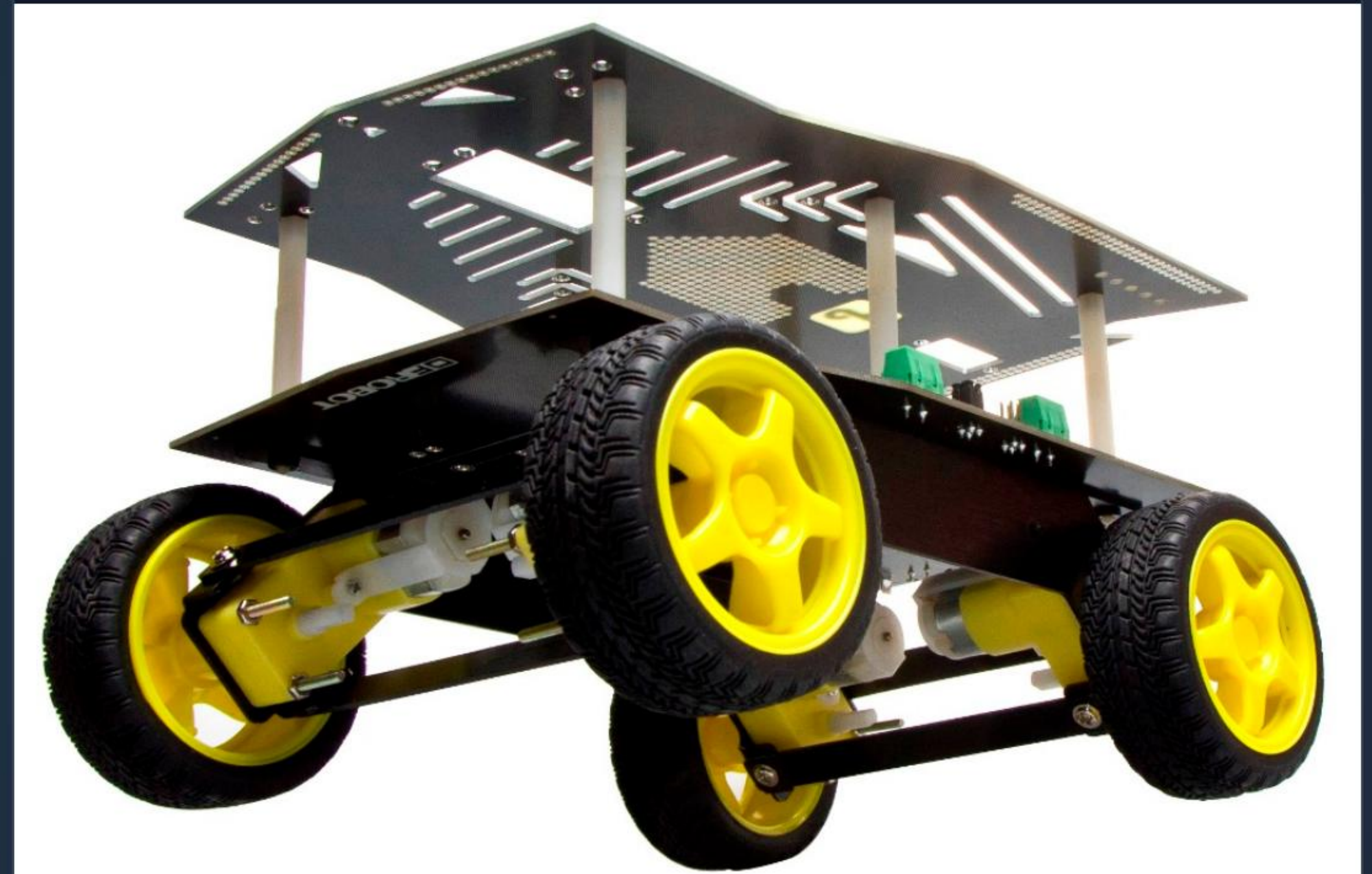
Navigation using Ultrasonic sensor & State Machine.

Goal: Navigate a labyrinth without human intervention.

Hardware Setup

-  **Chassis:** Cherokee 4WD + DC Motors
-  **Controller:** Arduino Uno + Motor Driver
-  **Sensor:** URM37 Ultrasonic (Servo mounted)
-  **Power:** 7.5V LiPo Battery

Note: Standard setup consistent with class requirements.



Initial Behavior

The Problem

-  Robot kept reversing repeatedly
-  Random, erratic turns
-  Never moved forward smoothly
-  Stuck in "Reverse → Turn → Reverse" loop



Root Cause: Timing Issue

Original Code

- Sensor readings called millions of times per second.
- No delays between readings.
- Control logic coupled tightly to noisy sensor data.

The Result

- Constant **false obstacle detection**.
- Motor control out of sync with sensor.
- Robot "thought" an obstacle was always present.

Solution: FSM + Timers

Finite State Machine (FSM)

- ↑ **FORWARD:** Scan & move
- ↓ **BACKWARD:** Reverse if obstacle
- ↻ **TURN:** Avoid obstacle
- ↺ **RESET:** Re-center servo

Metro Timers

Read sensor every **50ms** instead of continuously.

Decoupled sensor polling from motor control logic for precise timing.

Key Code Improvements

-  **Non-blocking delays:** Used `millis()` instead of `delay()`.
-  **Invalid Reading Filter:** Ignore noise (readings < 5cm).
-  **Servo Scan Range:** Increased from 60°–120° to **60°–150°**.
-  **Obstacle Threshold:** Reduced from 30cm to **15cm**.
-  **Speed Balancing:** Smoothed PWM values for motors.

Debugging Process

1

Step 1

Added Serial Print to monitor sensor values.

2

Observation

Readings happening too fast (1000+ Hz).

3

The Fix

Restored timer checks to **50ms**.

4

Retest

Verified smooth movement.

Lesson: Real-time systems need precise timing control.

Final Result

- ✓ Navigates labyrinth autonomously
- ✓ Maintains **~15cm** from obstacles
- ✓ Chooses turn direction based on last scan
- ✓ Recovers from dead ends

[Video Demo Placeholder]



Conclusion

Key Takeaways

- Timing is critical in embedded systems.
- State machines simplify complex behaviors.
- Serial monitor is essential for debugging.

Future Improvements

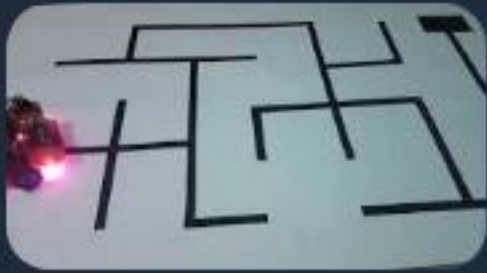
- Add **path memory** logic.
- Integrate multiple side sensors.
- Implement adaptive speed thresholds.

Image Sources



<https://raw.githubusercontent.com/DFRobot/DFRobotMediaWikiImage/master/Image/Rob0102hole.jpg>

Source: wiki.dfrobot.com



<https://external-preview.redd.it/2e4Blsvgb2xJjXGn5eW5FKcDqvVOfhbwskdaiZsc10.jpg?format=pjpg&auto=webp&s=746476e28106982f190870a9f943214a5d411b52>

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